AtHOL1 substrate expansion — benchmark report

Agent-alone candidates, published prediction values and reference variants calculated in this work.

Agent Scientist: gpt-6.1-sol. Predictor names and units are listed with each measurement.

Published predictions and our predictions

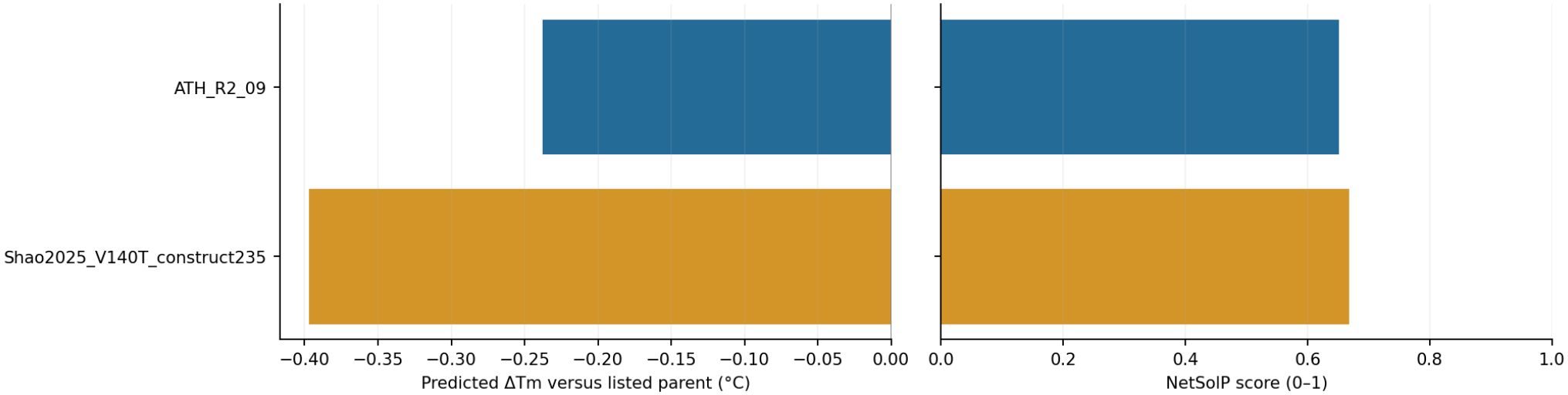
Property / predictor	Published prediction	Our candidates
Electrostatic stabilization estimate (kcal/mol)	WT −5.6; V140T −7.8	In progress: matched reaction-electrostatics assessment.
Projected electric field (MV/cm)	WT −35.2 ± 9.6; V140T −43.4 ± 5.7	In progress: matched reaction-electrostatics assessment.
Mean C–I / electric-field angle (degrees)	WT 41.2; V140T 16.6	In progress: matched reaction-electrostatics assessment.

Source: Shao et al., Figure 8A and electrostatic analysis

Property predictions calculated in this work

Sequence	Group	Parent / background	Predicted Tm (°C)	ΔTm (°C)	NetSolP	pHPred (raw)
ATH_R2_09	Our candidate	AtHOL1_3LCC_construct235	47.1996	-0.2375	0.651168	7.0714
Shao2025_V140T_construct235	Reference	AtHOL1_3LCC_construct235	47.0402	-0.3969	0.667472	7.0286

Candidate and reference property predictions



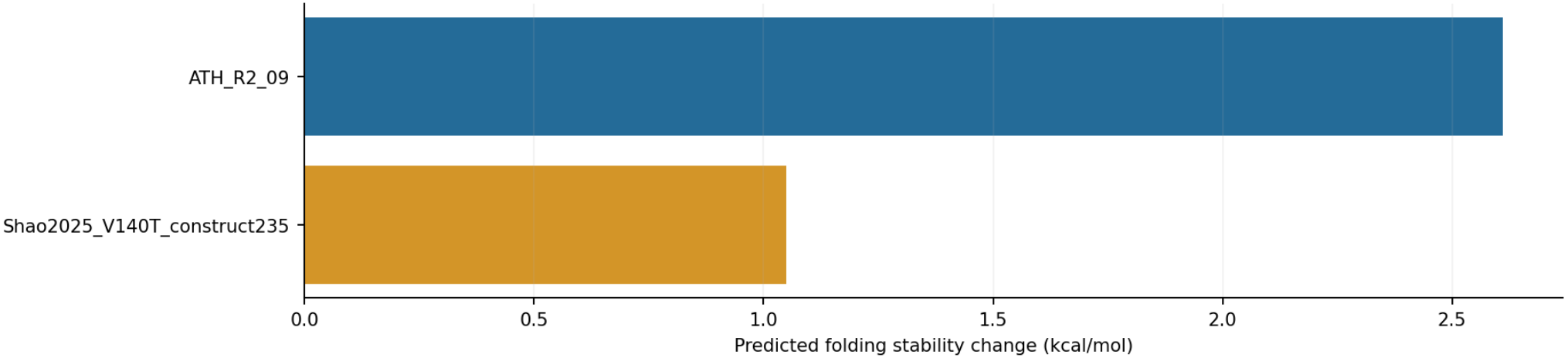
Blue: our candidates. Amber: reference variants calculated in this work. Changes use the parents listed in the table.

Predicted folding stability change — open-source assessment

Method: EvoEF1 1.1. Units: kcal/mol. Protocol: common WT; protein-only; 3 repair runs; 10 matched WT/mutant repacking runs; negative predicts stabilization. Values below were calculated in this work for the listed sequences.

Sequence	Group	Parent / background	Prediction (kcal/mol)
ATH_R2_09	Our candidate	AtHOL1_3LCC_construct235	2.61
Shao2025_V140T_construct235	Reported reference	AtHOL1_3LCC_construct235	1.05

Predicted folding stability change: reference and our sequences



Amber: reported reference variants recalculated here. Blue: our retained candidates. All rows in this panel use the stated method and protocol.

Parent identity, preparation details and run records: results/task_004_athol1_substrate/agent-alone/README.md.